DESIGN AND SIMULATION OF CURRENT-STARVED RING OSCILLATOR USING CADENCE VIRTUOSO

Frequency, Power, PVT considerations and Monte Carlo Analysis

Technical report

Tool Used: *Cadence Virtuoso (Schematic Editor, ADE-Explorer, ADE ASSEMBLER)*
Technology: *CMOS , 90nm Technology*

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1. Objective

The objective of this project is to **design and simulate a PMOS-source Current-Starved Ring Oscillator (CSRO) using 90 nm CMOS technology** and analyze its performance through different simulation techniques. The project focuses on **delay cell design and oscillation verification** followed by **frequency measurement and start-up analysis** to confirm the proper operation of the oscillator. The relationship between **oscillation frequency and control current/voltage** is studied to evaluate frequency tuning characteristics. In addition, the project aims to analyze **power consumption and PVT variations**, perform **Monte Carlo analysis** for process and mismatch variations, and complete the physical design through **layout, DRC, LVS, and PEX** to verify the correctness and post-layout performance of the designed oscillator .

2. Introduction to the Current-Starved Ring Oscillator

Ring Oscillator

A ring oscillator is a CMOS-based oscillator that generates a periodic output waveform without requiring an external clock or resonant components. It is formed by connecting an odd number of inverter stages in a closed loop, where the output of the final stage is fed back to the input of the first stage. Due to the inversion and propagation delay through each stage, the circuit continuously switches between logic states, producing a periodic oscillating waveform. The oscillation frequency mainly depends on the number of delay stages and the propagation delay of each stage.

Why a Current-Starved Ring Oscillator ?

A conventional ring oscillator has limited control over its oscillation frequency because the delay of each inverter is primarily determined by the transistor characteristics, supply voltage, load capacitance, and device dimensions. A **current-starved ring oscillator** provides better control of the oscillation frequency by limiting the current available to each inverter stage. In this project, the current-starving mechanism is implemented using **PMOS source transistors**, which control the charging current of the internal nodes.

The current-starved ring oscillator consists of multiple inverter-based delay stages connected in a feedback loop. The main functions of the circuit can be described as follows:

- **Delay stages:** Each inverter stage introduces a finite propagation delay to the signal passing through it.
- **PMOS current source:** The PMOS source limits the charging current available to the inverter stages, thereby controlling the delay of each stage.
- **Feedback connection:** The output of the final inverter is connected back to the input of the first inverter, allowing continuous switching and sustained oscillation.
- **Frequency control:** Changing the control voltage/current changes the available charging current and consequently changes the propagation delay and oscillation frequency.

The approximate oscillation frequency of an N-stage ring oscillator can be expressed as

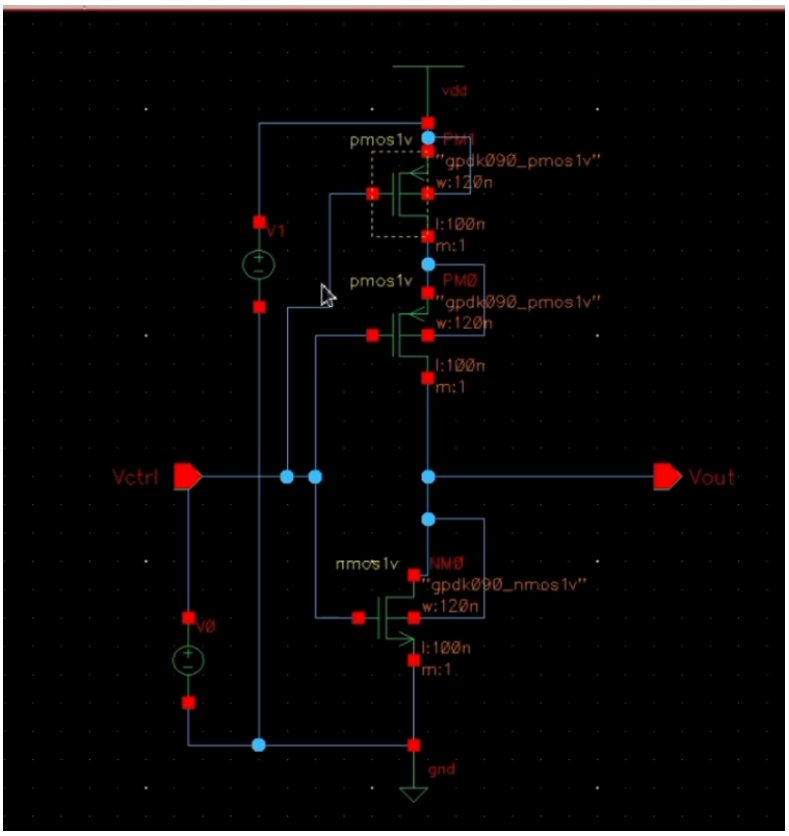
$$F_{osc} = \frac{1}{2}.N.t_d$$

where N is the number of delay stages and t_d is the average propagation delay of each stage.

In a current-starved configuration, the propagation delay is controlled by the limited current supplied to the inverter stage. Therefore, increasing or decreasing the available current changes the charging and discharging time of the internal capacitive nodes, resulting in a corresponding change in the oscillation frequency.

In this project, the **current-starved ring oscillator is designed and analyzed using 90 nm CMOS technology**, with the frequency-control mechanism implemented using **PMOS source devices**. The design is evaluated through transient simulation to verify start-up and steady-state oscillation, followed by frequency measurement and analysis of the relationship between the control condition and oscillation frequency.

Delay Cell Design



The basic delay cell of the proposed current-starved ring oscillator consists of three MOSF two PMOS transistors and one NMOS transistor.

- The upper PMOS acts as the current-starving/current-source transistor.Its gate is controlled by V_CTRL, which controls the amount of current supplied to the inverter.

- The middle PMOS and bottom NMOS form a CMOS inverter. The input is applied to their gates and the common drain connection provides the V_OUT.

- The controlled current from the upper PMOS determines how quickly the output node charges. Therefore, changing V_CTRL changes the available current and hence the propagation delay of the cell.

3. Functional Description of the Circuit

The proposed circuit is a **Current-Starved Ring Oscillator (CSRO)** designed using **90 nm CMOS technology**. The schematic consists of multiple CMOS inverter-based delay stages connected in a closed-loop configuration to generate a continuous oscillating signal. Each delay stage is formed using PMOS and NMOS transistors, while the **PMOS source transistors are used to control the current supplied to the inverter stages**. The control voltage Vctrl is applied to the current-starving PMOS devices, allowing the charging current of the internal nodes to be controlled.

When the circuit is powered by the supply voltage V_{DD}=1.8V, a small initial disturbance causes the inverter stages to switch between logic HIGH and LOW states. The signal propagates through the connected delay stages and is fed back to the input, resulting in continuous oscillation. The PMOS current-source devices restrict the available charging current, thereby increasing or decreasing the propagation delay depending on the applied control voltage. Consequently, the oscillation frequency can be controlled by varying Vctrl.

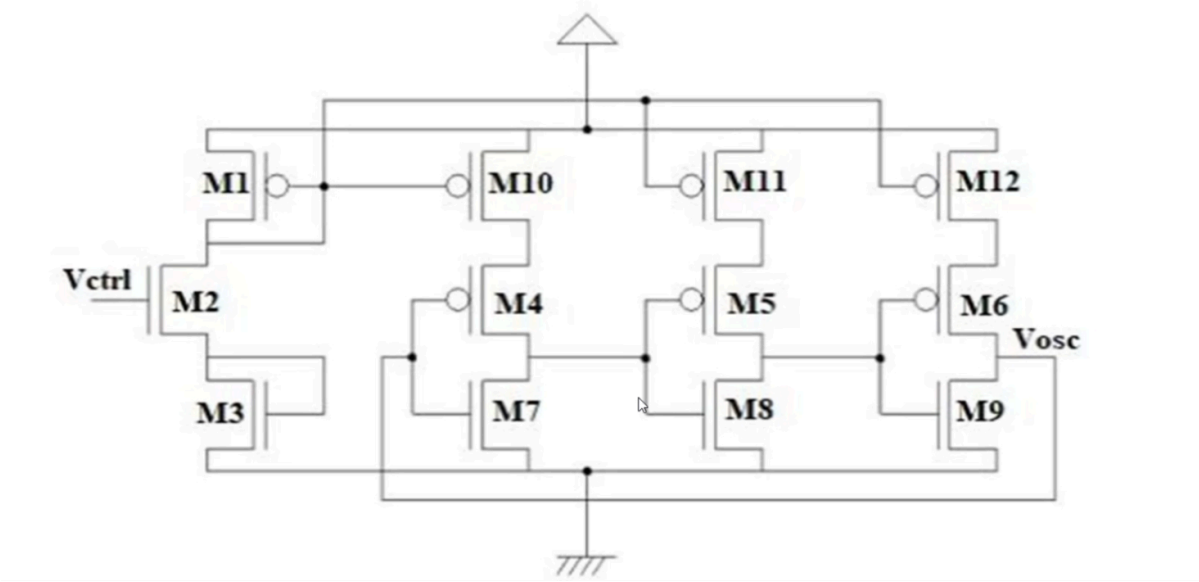
The circuit provides $\mathbf{V_{osc}}$ as the output, which represents the generated periodic waveform. The main objective of the circuit is to analyze the relationship between the **control voltage, propagation delay, and oscillation frequency**. By changing the control condition, the current supplied to the delay stages changes, resulting in a corresponding variation in the output frequency. Thus, the proposed current-starved ring oscillator provides a controllable oscillation source suitable for frequency-generation and timing-related applications.

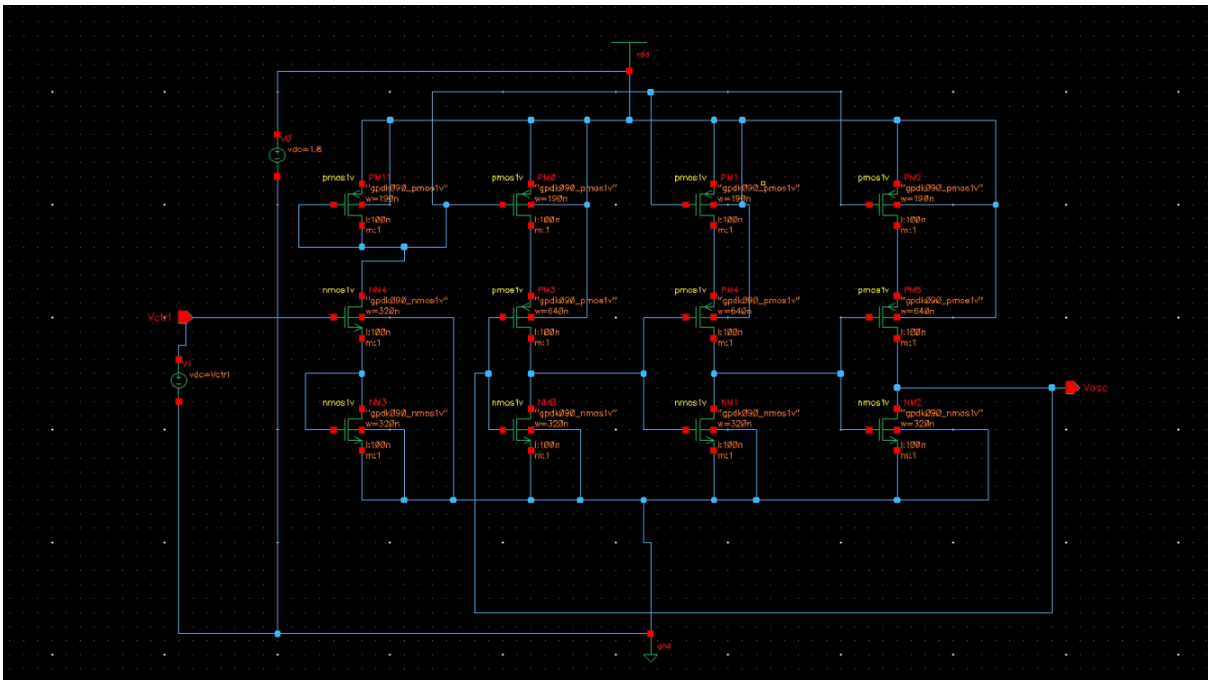
Main Circuit Functions

- **V_{DD}:** Provides the 1.8 V supply to the oscillator circuit.
- **V_{ctrl}:** Controls the PMOS current-source devices and hence the current available to the delay stages.
- **PMOS source devices:** Limit/control the charging current of the inverter stages.
- **CMOS inverter stages:** Provide the required signal inversion and propagation delay.
- **Feedback path:** Connects the final stage back to the input to sustain continuous oscillation.
- **Vosc:** Provides the final oscillating output waveform for transient analysis and frequency measurement.

The schematic of the designed **90 nm Current-Starved Ring Oscillator**

4. Circuit Architecture Implemented in Virtuoso





Schematic of the 3-Stage 90-nm PMOS Current-Starved Ring Oscillator

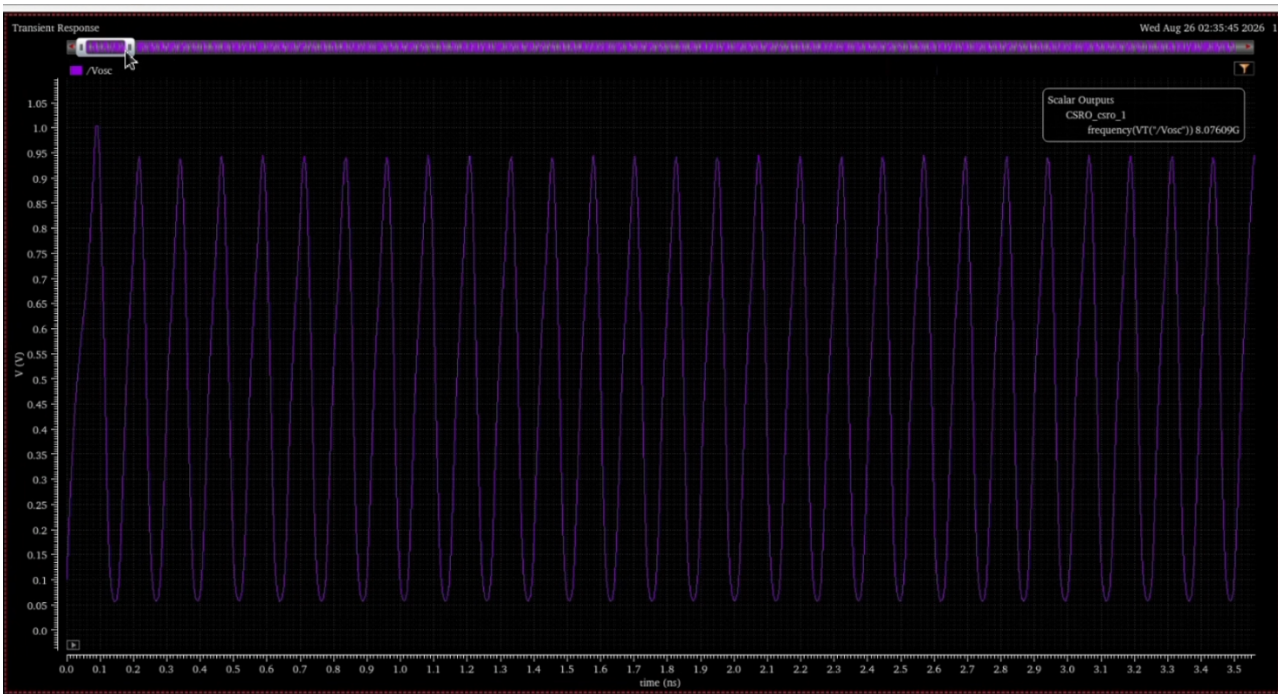
5. Simulation Setup

3-Stage PMOS Current-Starved Ring Oscillator

Simulation Conditions

Parameters	**Values**
Technology	90 nm CMOS
Circuit	3-Stage PMOS CSRO
Supply voltage VDD	1.8 V
Control voltage Vctrl	1 V
Analysis	Transient
Output node	Vosc
Frequency	Measured from output waveform

START-UP ANALYSIS



At the beginning of the simulation:

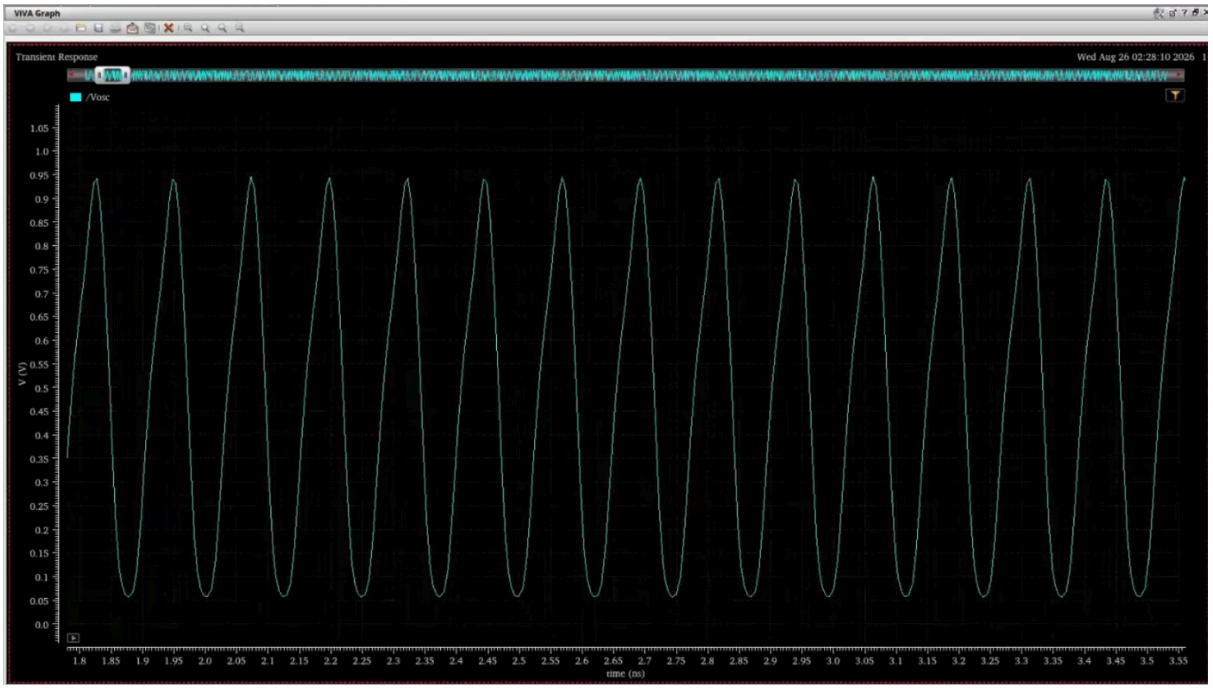
At t = 0, the waveform starts from an initial condition.

- The first oscillation has a larger peak (~1.0 V).

- After that, the waveform settles into a more consistent periodic oscillation, with peaks around ~0.94 V and valleys around ~0.06 V.

- After the initial transient, the waveform becomes stable and periodic.

6. Simulation Result – Output Waveform



The transient simulation waveform shown in the result demonstrates that the proposed **three-stage PMOS-current-source current-starved ring oscillator** successfully produces a periodic output signal. The waveform contains repeated rising and falling transitions, indicating sustained oscillation after circuit start-up.

Frequency Measurement

The oscillation frequency of the proposed **three-stage Current-Starved Ring Oscillator using PMOS current sources** was measured using transient simulation in Cadence Virtuoso. The output signal Vosc was analyzed using the frequency measurement function available in the calculator. For the simulation, the supply voltage was maintained at VDD=1.8V, and the control voltage was set to V_{ctrl} = 1V.

The frequency was calculated from the oscillation period of the output waveform using the relation:

$$F_{osc} = \frac{1}{T_1}$$

where Fosc is the oscillation frequency and T is the time period of one complete cycle of the output waveform.

The frequency measurement obtained from the Cadence simulation is:

f =8.076GHz

Thus, at $\mathbf{V_{DD}}$**=1.8V** and $\mathbf{V_{ctrl}}$**=1V**, the proposed PMOS current-source current-starved ring oscillator achieves an oscillation frequency of approximately **8.076 GHz**. This result confirms that the designed three-stage oscillator successfully generates a high-frequency periodic output signal.

t_1 =1.014ns

t_2 =1.138ns

Finding value of T

T=1.138-1.014

T=0.124ns

Frequency =1 / 0.124ns

f=8.065GHz

3-stage CSRO: t(d) value

Formula

T=2Nt(d)

N=3 (Number of stages)

t(d)=T\6

For VCTRL=1.0V

t(d)=0.124ns\6

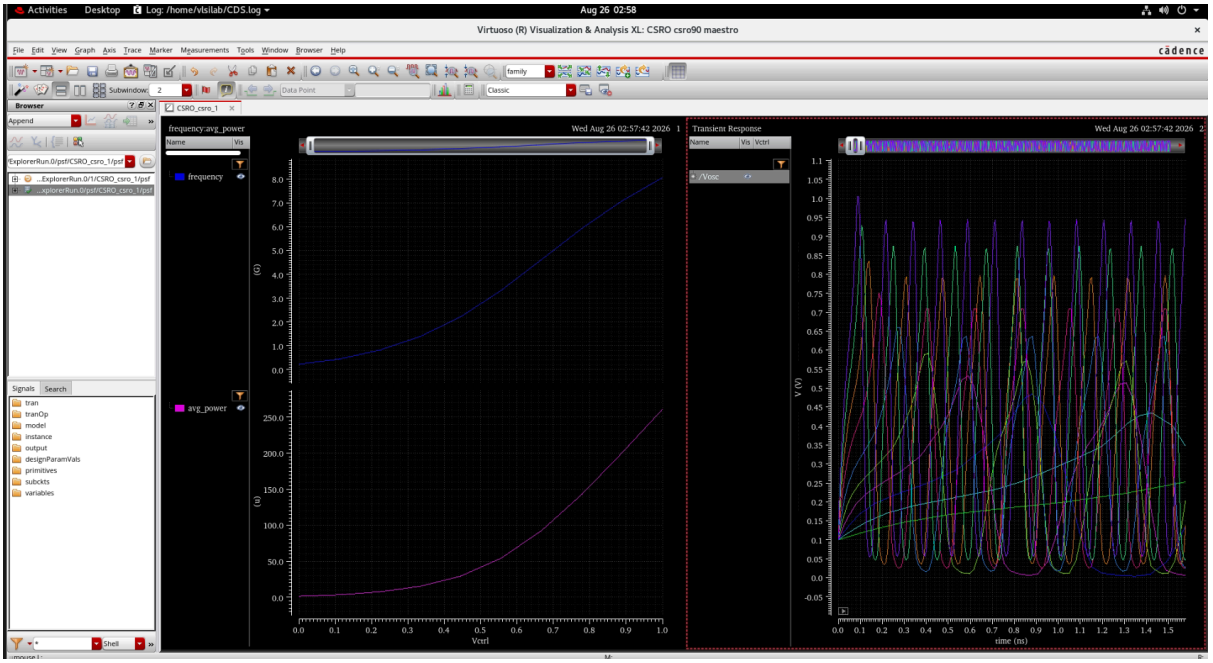
t(d)=0.020667ns = 20.667ps

FREQUENCY MEASUREMENT :-

f=1/2(3)(20.667ps)

f ≈ 8.065GHz

Frequency and Power Analysis with Respect to Control Voltage



A parametric analysis was performed to investigate the effect of the control voltage Vctrl on the performance of the proposed **three-stage Current-Starved Ring Oscillator using PMOS current sources**. The V_{ctrl} voltage was varied from **0 V to 1 V**, while the supply voltage was maintained at V_{DD}=1.8V. The variation of oscillation frequency and average power with respect to the control voltage was obtained using Cadence Virtuoso.

Frequency versus Control Voltage

The upper-left graph shows the variation of the **oscillation frequency with V_{ctrl}**. It can be observed that the oscillation frequency increases as the control voltage increases.

At lower control voltages, the PMOS current-source devices provide a smaller effective current to the inverter stages. This increases the charging and discharging time of the internal nodes and therefore increases the propagation delay. Since the oscillation frequency is inversely proportional to the propagation delay,

$$F_{osc} = \frac{1}{2}.N.t_p$$

The oscillator operates at a lower frequency.

As V_{ctrl} increases, the available current increases, reducing the propagation delay of the stages. Consequently, the oscillation frequency increases. From the simulation, the frequency reaches approximately **8.1 GHz at V_{ctrl}=1V**. This demonstrates the voltage-controlled frequency characteristic of the proposed CSRO.

Average Power versus Control Voltage

The lower-left graph shows the variation of **average power consumption with V_{ctrl}**. The average power increases as the control voltage increases.

This occurs because increasing V_{ctrl} changes the current supplied by the PMOS current-source devices, allowing greater current to flow through the oscillator stages. The increased current results in higher power consumption during circuit operation.

From the simulation graph, the average power increases to approximately **260µW at V_{ctrl}=1V**. Therefore, there is a trade-off between oscillator speed and power consumption: increasing V_{ctrl} provides a higher oscillation frequency but also results in higher power consumption.

Transient Response for Different Control Voltages

The right-side graph presents the **transient output response V_{osc}** for different values of the control voltage. The waveforms demonstrate that the oscillator produces continuous periodic oscillations after start-up.

As V_{ctrl} is increased, the oscillation period becomes smaller, corresponding to an increase in oscillation frequency. At lower V_{ctrl}, the oscillations are slower because the current-starving effect is stronger and the propagation delay is higher. At higher V_{ctrl}, the oscillations occur more rapidly due to the increased current available to the inverter stages.

This transient analysis confirms the expected **voltage-controlled oscillation behavior** of the proposed circuit.

Overall Observation

The simulation results demonstrate that the proposed **90 nm three-stage PMOS current-source Current-Starved Ring Oscillator** provides effective frequency control through V_{ctrl}. The oscillation frequency increases with

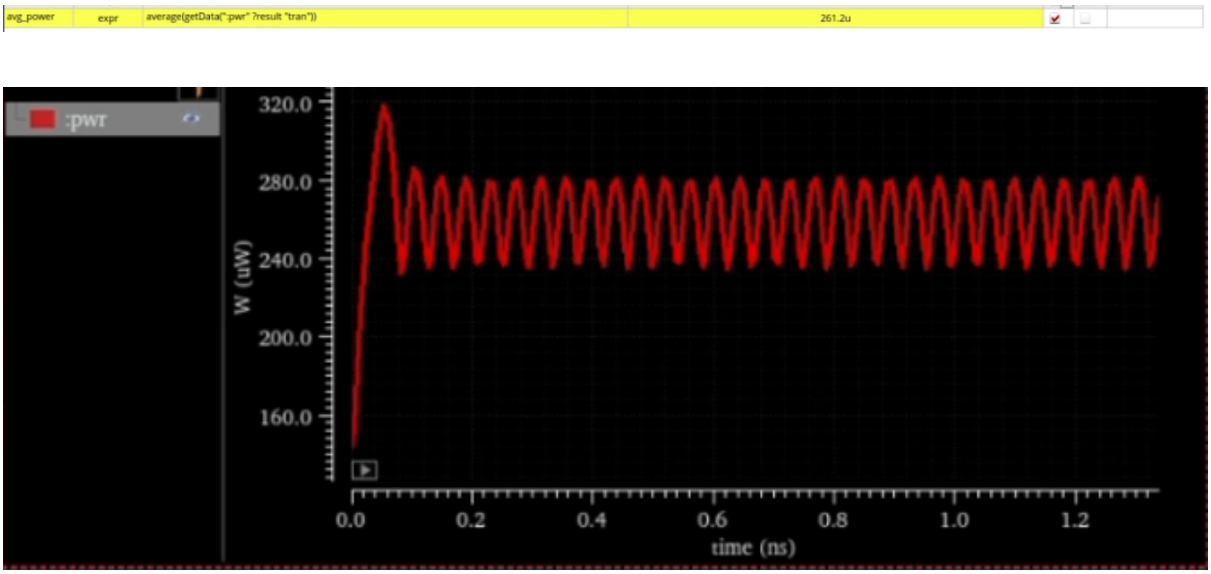
increasing control voltage, while the average power consumption also increases. Therefore, V_{ctrl} can be selected according to the required balance between **operating frequency and power consumption**.

At the maximum simulated control voltage of **V_{ctrl}=1V**, the oscillator achieves approximately **8.1 GHz** frequency with approximately **260 µW** average power consumption.

Table for your report

Vctrl	Frequency	Average Power
0 V	Low	Low
0.2 V	Increasing	Increasing
0.4 V	Increasing	Increasing
0.6 V	Increasing	Increasing
0.8 V	Increasing	Increasing
1.0 V	8.1 GHz	260 µW

AVG POWER VALUE AND PWR GRAPH



7. PVT Analysis

PVT (Process, Voltage, and Temperature) analysis is performed to evaluate the robustness and performance of the proposed **90 nm Current-Starved Ring Oscillator with PMOS current sources** under different process, voltage/control conditions, and temperature variations. The purpose of this analysis is to determine how changes in transistor process corners and temperature affect the oscillation frequency and power consumption of the ring oscillator. The simulations are carried out for five control-voltage values, namely **0.5V, 0.625V, 0.75V, 0.875V, and 1V**, while considering the process corners **FF, FS, NN, SF, and SS** at temperatures of **–40°C, 27°C, and 125°C**.

Frequency Variation with PVT

The frequency response obtained from the PVT analysis is shown in the frequency-versus-temperature plot. From the graph, it can be observed that the oscillation frequency generally **decreases as temperature increases**. This occurs because temperature variation changes the electrical characteristics and carrier mobility of the MOS transistors, which affects the charging and discharging current of the delay stages. Since the proposed circuit is current-starved, the available current directly controls the propagation delay of each stage and consequently determines the oscillation frequency.

The frequency also changes significantly with the control voltage. As $\mathbf{V_{ctrl}}$ **increases, the available current through the PMOS current-source arrangement increases**, reducing the delay of each stage and resulting in a higher oscillation frequency. The simulation results therefore demonstrate the expected frequency-control characteristic of the proposed current-starved ring oscillator.

From the obtained PVT results, the frequency ranges from approximately **1.238 GHz to 13.87 GHz** for the different control-voltage values, process corners, and temperature conditions. At $\mathbf{V_{ctrl}}$**=1V**, the maximum frequency observed in the table is approximately **13.87 GHz**, while at $\mathbf{V_{ctrl}}$ **= 0.5V**, the maximum frequency is approximately **5.118 GHz**. These results confirm that the oscillator provides a wide frequency tuning range.

Power Variation with PVT

The power-versus-temperature graph shows the variation of power consumption for different process corners and control-voltage values. It can be observed that the power consumption generally **decreases with increasing temperature** for the simulated conditions. The power also increases as $\mathbf{V_{ctrl}}$ **is increased**, because a higher control voltage results in a larger current through the current-starved stages. Since dynamic power is related to switching activity and operating frequency, the increase in oscillation frequency contributes to higher power consumption.

From the numerical PVT results, the power consumption varies approximately from **19.24 µW to 469.9 µW** across the different process corners, temperatures, and control-voltage conditions. At $\mathbf{V_{ctrl}}$ **= 1 V**, the maximum power obtained is approximately **469.9 µW**, whereas at $\mathbf{V_{ctrl}}$ **= 0.5 V**, the maximum value is approximately **75.15 µW**. Therefore, the simulation confirms the trade-off between **higher operating frequency and increased power consumption**.

Process-Corner Analysis

The PVT simulation considers five process corners:

- **FF – Fast NMOS, Fast PMOS**
- **FS – Fast NMOS, Slow PMOS**
- **NN – Nominal NMOS, Nominal PMOS**
- **SF – Slow NMOS, Fast PMOS**
- **SS – Slow NMOS, Slow PMOS**

Each process corner is evaluated at **-40°C, 27°C, and 125°C**. The difference between the process corners results from variations in transistor characteristics caused by manufacturing-process variations. Since the oscillator depends on the charging and discharging capability of the MOS transistors, these variations directly influence both frequency and power consumption.

The PVT table shows that the **FF corner generally provides higher frequency and power**, while the **SS corner generally produces lower frequency and power**. The FS and SF corners provide intermediate behavior depending on the relative speed of the NMOS and PMOS devices. This confirms that the proposed oscillator is sensitive to process variations, as expected for a CMOS current-controlled oscillator.

Temperature Analysis

The temperature sweep is performed over **-40°C, 27°C, and 125°C**. The frequency plots show a decreasing trend with temperature for the different process corners. For example, at $\mathbf{V_{ctrl}}$ **= 1 V**, the frequency for the FF corner changes from approximately **13.87 GHz at -40°C** to **9.969 GHz at 27°C** and approximately **7.157 GHz at 125°C**. Similarly, the power decreases from approximately **469.9 µW at -40°C** to **305.3 µW at 27°C** and approximately **226.1µW at 125°C** for the corresponding FF condition.

This temperature-dependent behavior demonstrates that the proposed oscillator maintains oscillation over the investigated temperature range, although its frequency and power characteristics vary with temperature.

PVT Transient Response

The transient-response plot contains the oscillator output waveforms for different combinations of process corners, temperatures, and control voltages. The waveforms confirm that the circuit successfully starts oscillating and produces a periodic output signal under the investigated PVT conditions. The variation in the period of the waveforms indicates the change in oscillation frequency caused by process and temperature variations.

The transient simulations therefore verify the functionality of the proposed **three-stage current-starved ring oscillator with PMOS current sources** under different PVT conditions.

PVT Results Summary

Parameter	Observed Result
Technology	90 nm CMOS
Supply voltage, VDD	1.8 V
Control voltage range	0.5–1 V
Temperature range	-40°C to 125°C
Process corners	FF, FS, NN, SF, SS
Maximum frequency	**13.87 GHz**
Minimum frequency	**1.238 GHz**
Maximum power	**469.9 µW**
Minimum power	**19.24 µW**

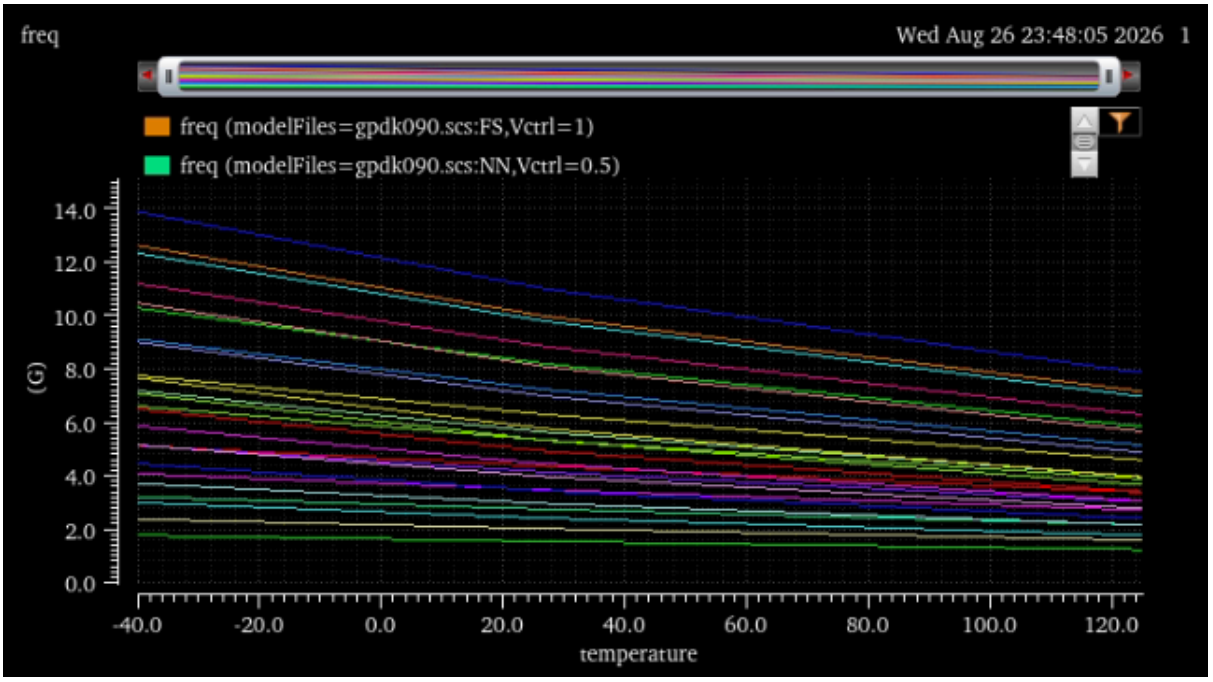
		Parameter					C0	C1	C2	C3	C4	C5
		gpdk090.scs					FF	FF	FF	FS	FS	FS
		temperature					-40	27	125	-40	27	125

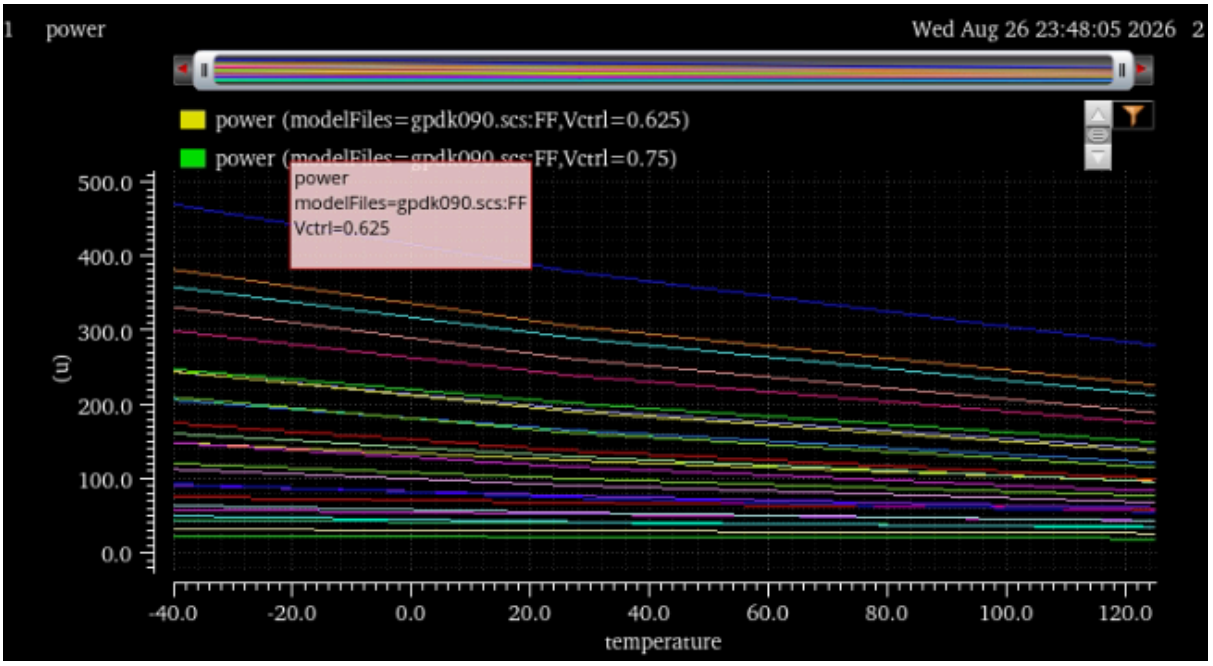
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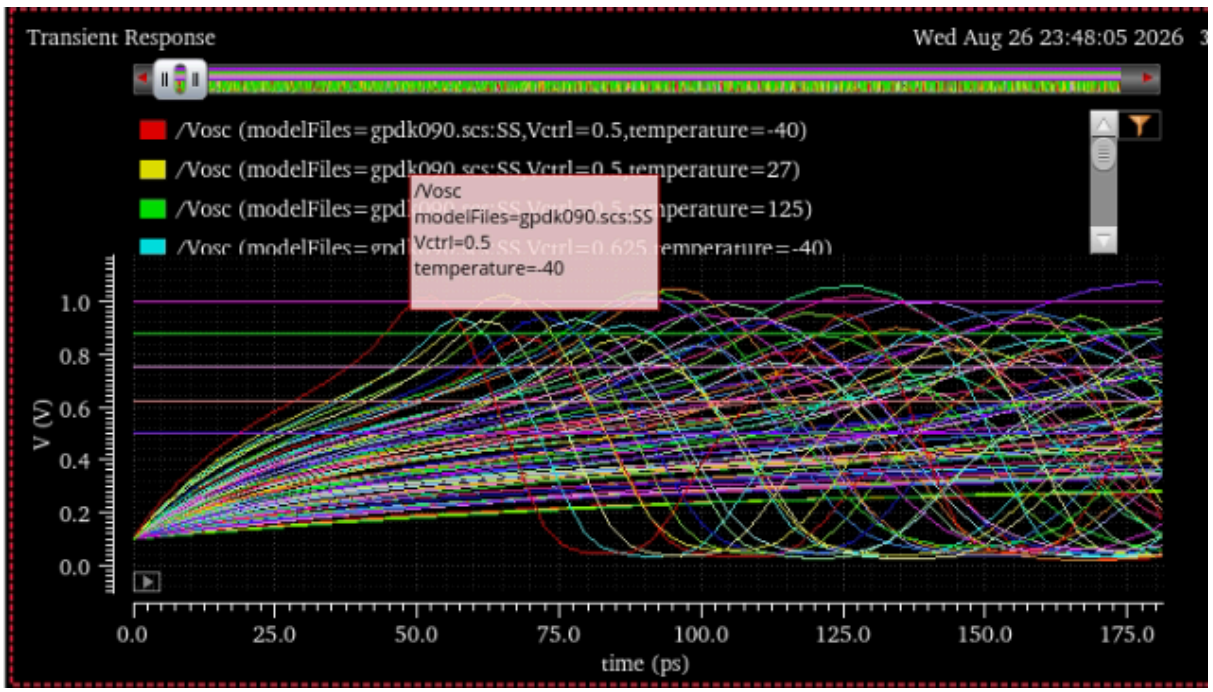
Point	Test	Output	Spec	Weight	Pass/Fail	Min	Max	C0	C1	C2	C3	C4	C5
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
Parameters: Vctrl=1													
1	CSRO_csro_1	Vout											
1	CSRO_csro_1	freq				3.658G	13.87G	13.87G	10.99G	7.847G	12.61G	9.969G	7.157G
1	CSRO_csro_1	power				115u	469.9u	469.9u	379.1u	279.6u	381.7u	305.3u	226.1u
1	CSRO_csro_1	/Vctrl											
Parameters: Vctrl=875m													
2	CSRO_csro_1	Vout											
2	CSRO_csro_1	freq				3.056G	12.32G	12.32G	9.78G	6.955G	11.18G	8.838G	6.287G
2	CSRO_csro_1	power				82.76u	358.4u	358.4u	289.8u	212.6u	298.9u	239u	174.3u
2	CSRO_csro_1	/Vctrl											
Parameters: Vctrl=750m													
3	CSRO_csro_1	Vout											
3	CSRO_csro_1	freq				2.405G	10.27G	10.27G	8.192G	5.837G	9.108G	7.227G	5.139G
3	CSRO_csro_1	power				54.83u	247.2u	247.2u	201.5u	149.3u	205.4u	165.4u	121.5u
3	CSRO_csro_1	/Vctrl											
Parameters: Vctrl=625m													
4	CSRO_csro_1	Vout											
4	CSRO_csro_1	freq				1.771G	7.747G	7.747G	6.291G	4.589G	6.601G	5.33G	3.881G
4	CSRO_csro_1	power				33.47u	148.3u	148.3u	124.4u	96.21u	120.3u	99.88u	76.56u
4	CSRO_csro_1	/Vctrl											
Parameters: Vctrl=500m													
5	CSRO_csro_1	Vout											
5	CSRO_csro_1	freq				1.238G	5.118G	5.118G	4.351G	3.378G	4.074G	3.47G	2.714G
5	CSRO_csro_1	power				19.24u	75.15u	75.15u	67.65u	57.49u	57.77u	51.9u	44u
5	CSRO_csro_1	/Vctrl											

C6	C7	C8	C9	C10	C11	C12	C13	C14
NN	NN	NN	SF	SF	SF	SS	SS	SS
-40	27	125	-40	27	125	-40	27	125

C6	C7	C8	C9	C10	C11	C12	C13	C14
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
10.46G	8.08G	5.657G	7.646G	5.756G	3.953G	7.072G	5.325G	3.658G
331.4u	261.2u	188.9u	244.6u	190.1u	136.3u	209.7u	161.6u	115u
8.993G	6.977G	4.874G	6.488G	4.921G	3.394G	5.853G	4.442G	3.056G
244u	193.4u	140u	174.6u	137.5u	99.68u	148.2u	115.5u	82.76u
7.177G	5.625G	3.962G	5.144G	3.966G	2.778G	4.443G	3.432G	2.405G
160.7u	129.4u	95.57u	112.5u	90.78u	67.98u	92.75u	74.07u	54.83u
5.128G	4.137G	3.018G	3.704G	2.959G	2.158G	2.992G	2.411G	1.771G
91.24u	76.63u	59.96u	63.61u	54.06u	43.33u	49.6u	41.97u	33.47u
3.208G	2.745G	2.165G	2.366G	2.018G	1.588G	1.768G	1.535G	1.238G
44.04u	40.29u	35.14u	31.53u	29.23u	26.11u	22.83u	21.36u	19.24u


freq
Wed Aug 26 23:48:05 2026 1
freq (modelFiles=gpdk090.scs:FS,Vctrl=1)
freq (modelFiles=gpdk090.scs:NN,Vctrl=0.5)
14.0
12.0
10.0
8.0
6.0
4.0
2.0
0.0
(G)
-40.0
-20.0
0.0
20.0
40.0
60.0
80.0
100.0
120.0
temperature





The PVT analysis demonstrates that the proposed 90 nm three-stage current-starved ring oscillator with PMOS current sources operates successfully over different process corners and temperatures. The oscillator provides a wide frequency tuning range while maintaining operation from -40°C to 125°C. The results also demonstrate the expected trade-off between frequency and power consumption: increasing Vctrl increases the available current, thereby increasing oscillation frequency but also increasing power consumption. Thus, the PVT simulations verify the functionality and

robustness of the proposed oscillator under process and temperature variations.

8. Monte Carlo Analysis

Monte Carlo analysis is performed to evaluate the statistical variation and robustness of the proposed **90 nm three-stage Current-Starved Ring Oscillator with PMOS current sources**. Unlike PVT analysis, which evaluates predefined process corners and temperature conditions, Monte Carlo analysis considers random variations in device parameters and evaluates their effect on the circuit performance. This analysis is useful for determining whether the oscillator can maintain its required frequency and power characteristics in the presence of manufacturing variations.

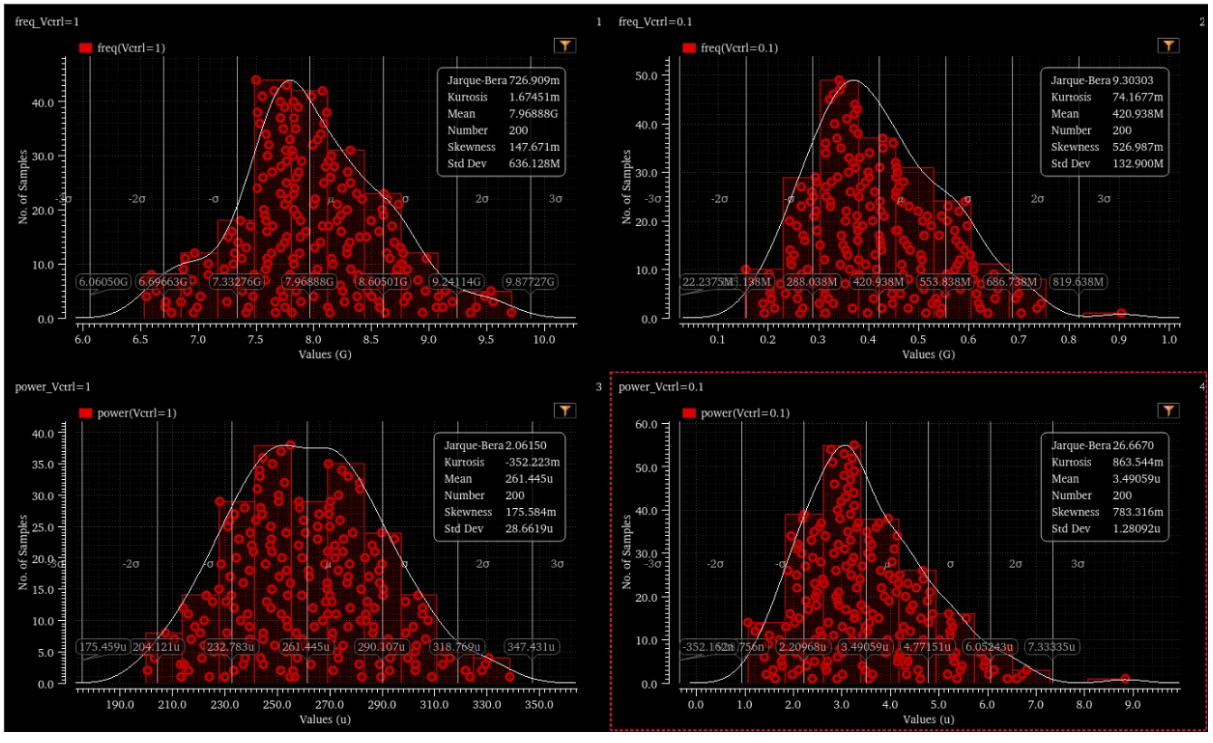
The Monte Carlo simulation was carried out in **Cadence Virtuoso** using **200 simulation samples**. The analysis was performed for two different control-voltage conditions, $\mathbf{V_{ctrl}}$ **= 1V** and $\mathbf{V_{ctrl}}$ **= 0.1V**. The output frequency and power consumption were selected as the main performance parameters for statistical analysis. The obtained results are represented using histograms along with fitted statistical distributions.

Monte Carlo Analysis at V_{ctrl} = 1 V

For $\mathbf{V_{ctrl}}$ **= 1 V**, the Monte Carlo frequency distribution is shown in the corresponding histogram. The simulation was performed for **200 samples**, and all 200 samples successfully produced valid results, giving a **100% yield**.

The frequency distribution has a **mean frequency of 7.96888 GHz** with a **standard deviation of 636.128 MHz**. The obtained frequency values range approximately from **6.528 GHz to 9.715 GHz**. The histogram shows that most of the samples are concentrated

Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Overall Yield Estimate: 100 %(200 passed/200 pts) Confidence Level: <not set> Filter: <not set>									
Yield Estimate: 100 %(200 passed/200 pts) | Parameters: Vctrl=1									
	CSRO_csro_1								
	freq(summary)	100% (200/200)	6.528G	info	9.715G	7.969G	636.1M		0
	freq	100% (200/200)	6.528G	info	9.715G	7.969G	636.1M		0
	power(summary)	100% (200/200)	199.8u	info	338.7u	261.4u	28.66u		0
	power	100% (200/200)	199.8u	info	338.7u	261.4u	28.66u		0
Yield Estimate: 100 %(200 passed/200 pts) | Parameters: Vctrl=0.1									
	CSRO_csro_1								
	freq(summary)	100% (200/200)	154.8M	info	903.7M	420.9M	132.9M		0
	freq	100% (200/200)	154.8M	info	903.7M	420.9M	132.9M		0
	power(summary)	100% (200/200)	1.062u	info	8.839u	3.491u	1.281u		0
	power	100% (200/200)	1.062u	info	8.839u	3.491u	1.281u		0



around the mean value, indicating that the oscillator maintains a relatively stable frequency despite random variations.

The Monte Carlo power analysis at $\mathbf{V_{ctrl}}$ **= 1 V** gives a **mean power consumption of 261.445 µW** with a **standard deviation of 28.6619 µW**. The power values obtained from the 200 samples range approximately from **199.8 µW to 338.7 µW**. The distribution indicates that the power consumption varies around the nominal operating condition because of device parameter variations.

Therefore, at $\mathbf{V_{ctrl}}$ **= 1 V**, the proposed oscillator provides a high operating frequency while maintaining a reasonably controlled statistical variation in both frequency and power.

Monte Carlo Analysis at V_{ctrl} = 0.1 V

The Monte Carlo analysis was also performed at a lower control voltage of $\mathbf{V_{ctrl}}$**=0.1 V**. Similar to the previous case, **200 samples** were simulated and all samples successfully passed the defined simulation conditions, resulting in a **100% yield**.

For frequency, the obtained mean value is approximately **420.938 MHz**, with a standard deviation of **132.900 MHz**. The frequency varies from approximately **154.8 MHz to 903.7 MHz** across the 200 Monte Carlo samples. The larger relative spread compared with the V_{ctrl} = 1 V case indicates that the oscillator frequency becomes more sensitive to device variations when the available current is low.

For power consumption, the mean value is approximately **3.49059 µW**, with a standard deviation of **1.28092 µW**. The observed power values range from approximately **1.062 µW to 8.839 µW**. The significantly lower power compared with the V_{ctrl}=1 V condition is due to the reduced current available to the current-starved stages.

Monte Carlo Results

Parameter	Vctrl = 1 V	Vctrl = 0.1 V
Number of samples	200	200
Yield	100%	100%
Mean frequency	**7.96888 GHz**	**420.938 MHz**
Frequency Std. Dev.	**636.128 MHz**	**132.900 MHz**
Minimum frequency	**6.528 GHz**	**154.8 MHz**
Maximum frequency	**9.715 GHz**	**903.7 MHz**
Mean power	**261.445 µW**	**3.49059 µW**
Power Std. Dev.	**28.6619 µW**	**1.28092 µW**
Minimum power	**199.8 µW**	**1.062 µW**
Maximum power	**338.7 µW**	**8.839 µW**

Interpretation of the Monte Carlo Results

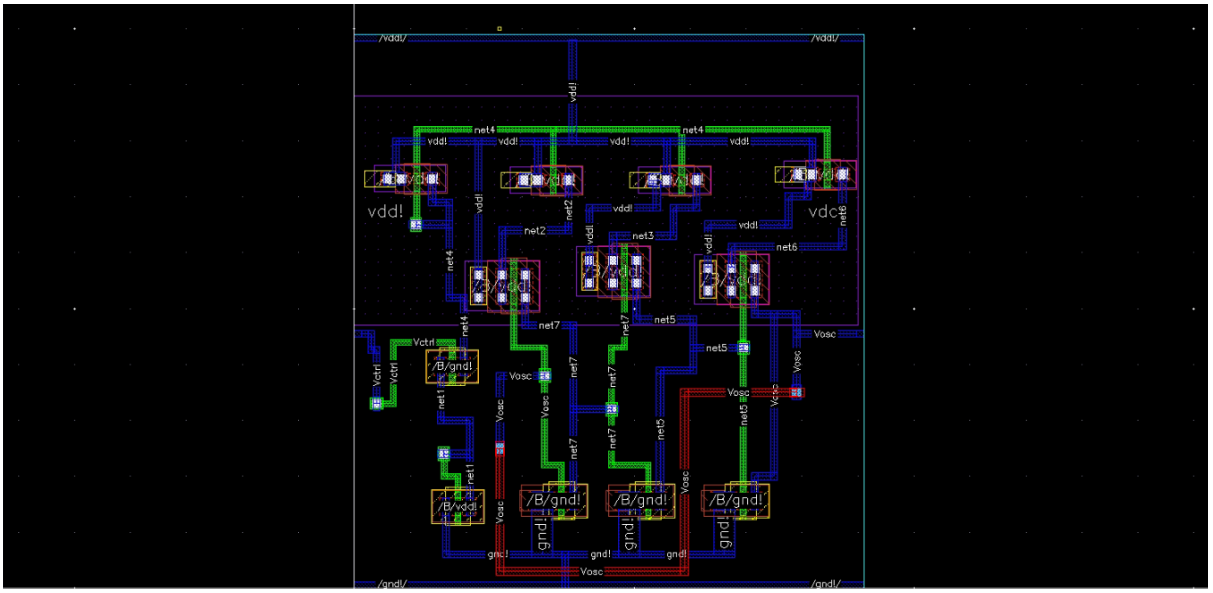
The Monte Carlo results demonstrate that the proposed current-starved ring oscillator is capable of maintaining oscillation under random device variations. The **100% yield for all 200 samples** at both control-voltage conditions indicates that every simulated sample produced a valid result under the selected test conditions.

At $\mathbf{V_{ctrl}}$ **= 1 V**, the oscillator achieves a mean frequency of approximately **7.97GHz**, whereas reducing the control voltage to **0.1 V** reduces the mean frequency to approximately **420.94 MHz**. This confirms the current-starving principle of the proposed circuit: **the control voltage regulates the available current, which controls the propagation delay of each inverter stage and hence the oscillation frequency**.

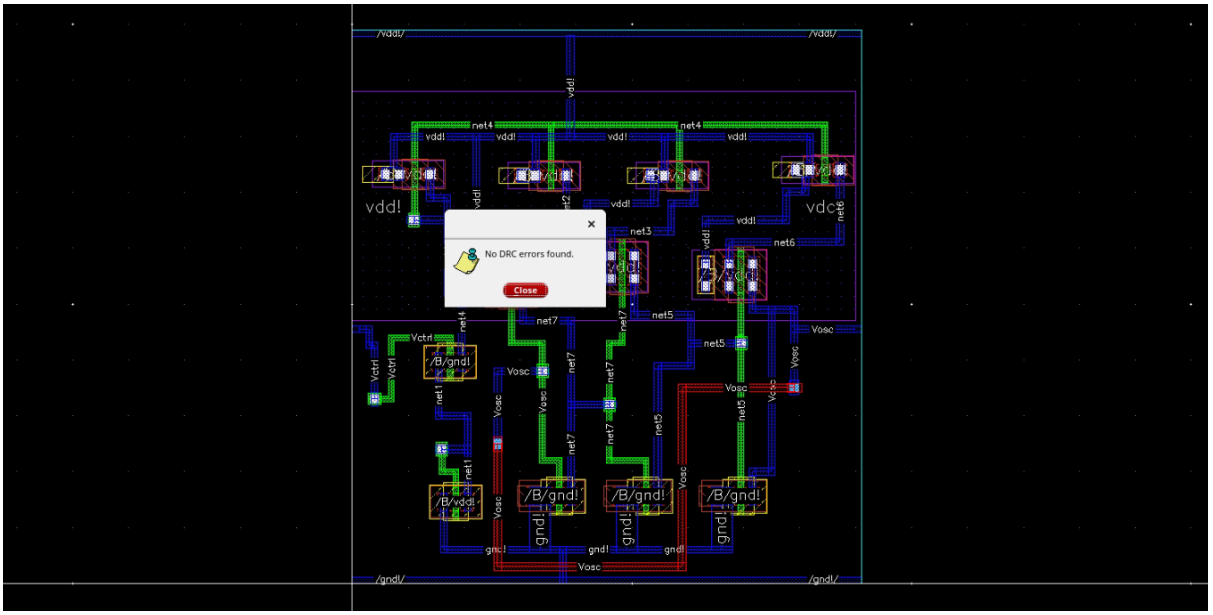
The power consumption also changes significantly with V_{ctrl}. At **1 V**, the mean power is approximately **261.45 µW**, while at **0.1 V**, it decreases to approximately **3.49 µW**. Thus, the lower-control-voltage condition provides substantially lower power consumption at the expense of a lower operating frequency.

The histogram plots also show the spread of the frequency and power values around their respective mean values. The standard deviation provides a measure of the sensitivity of the circuit to random variations. Overall, the Monte Carlo analysis confirms the **statistical robustness and functionality** of the proposed 90 nm current-starved ring oscillator and demonstrates the trade-off between **frequency tuning and power consumption**.

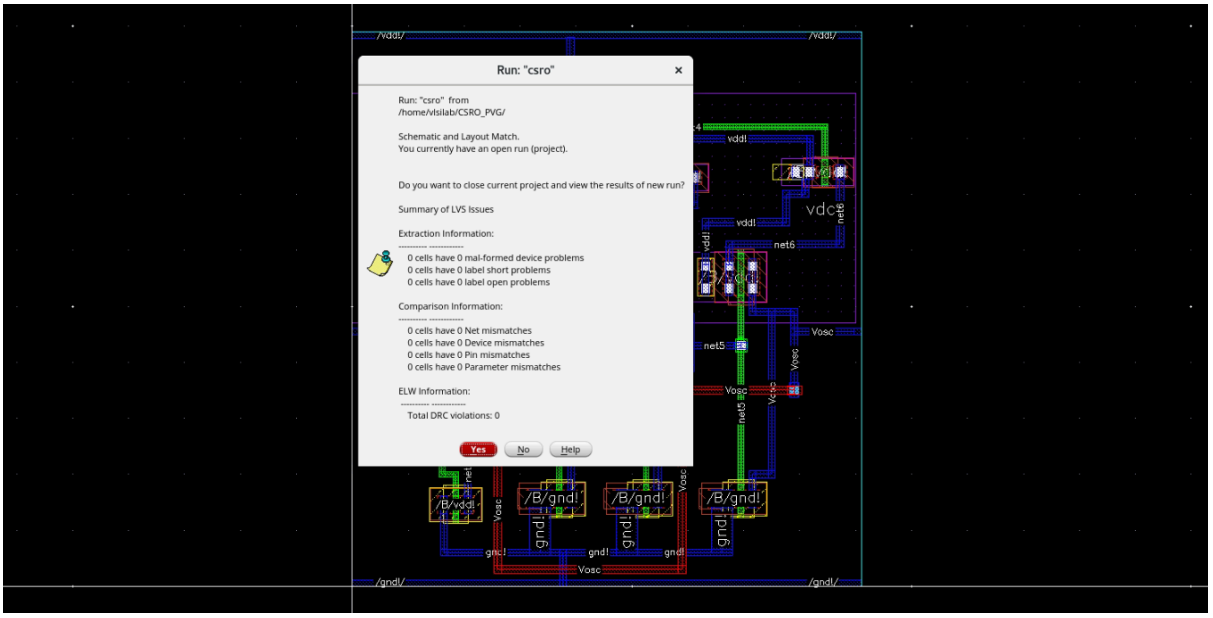
9. LAYOUT DESIGN

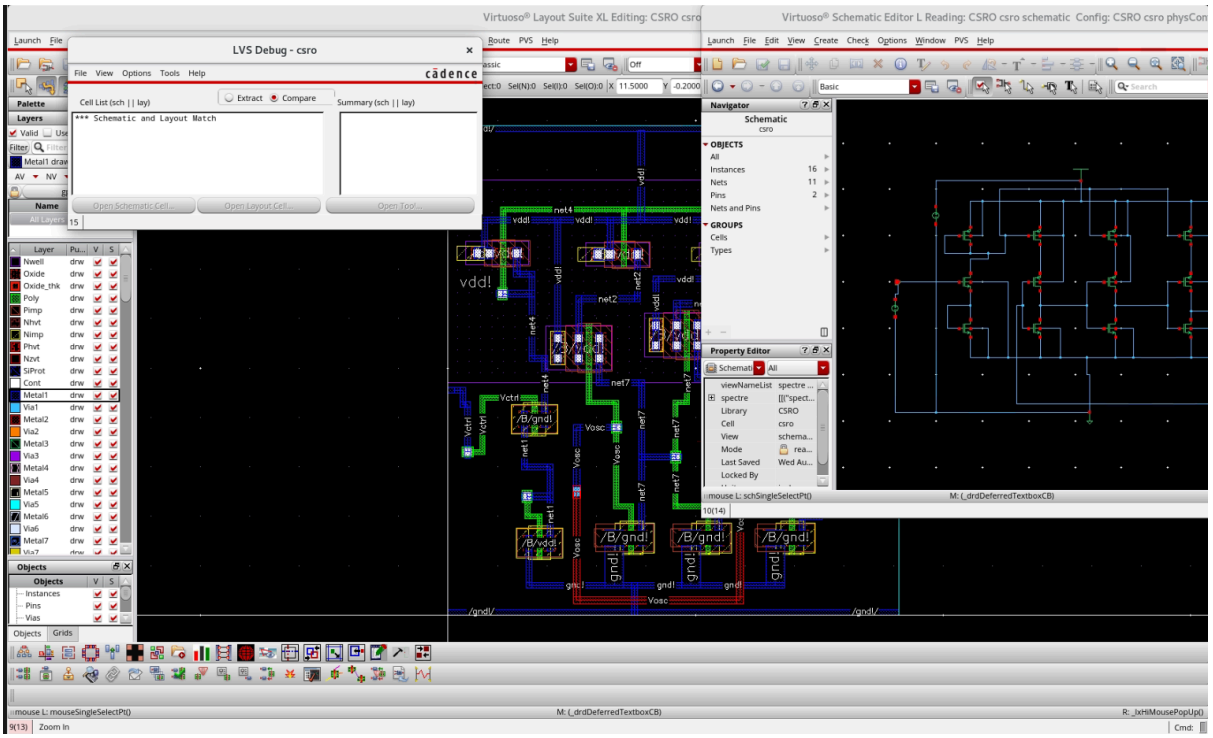


DRC CHECK

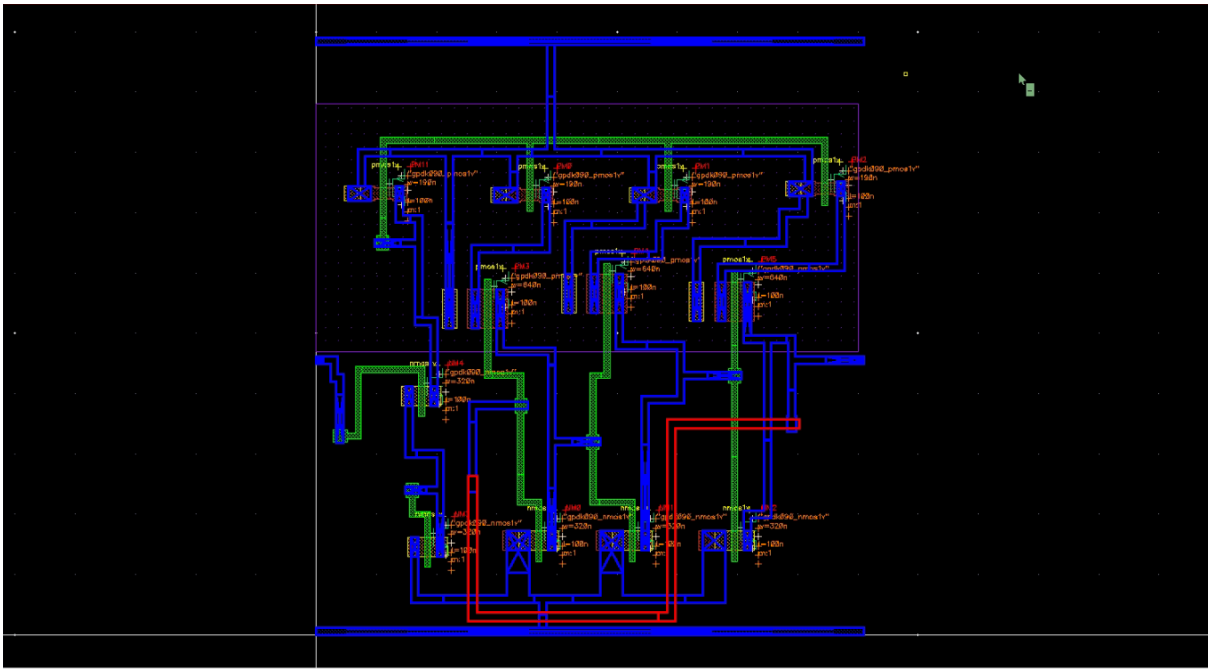


LVS CHECK

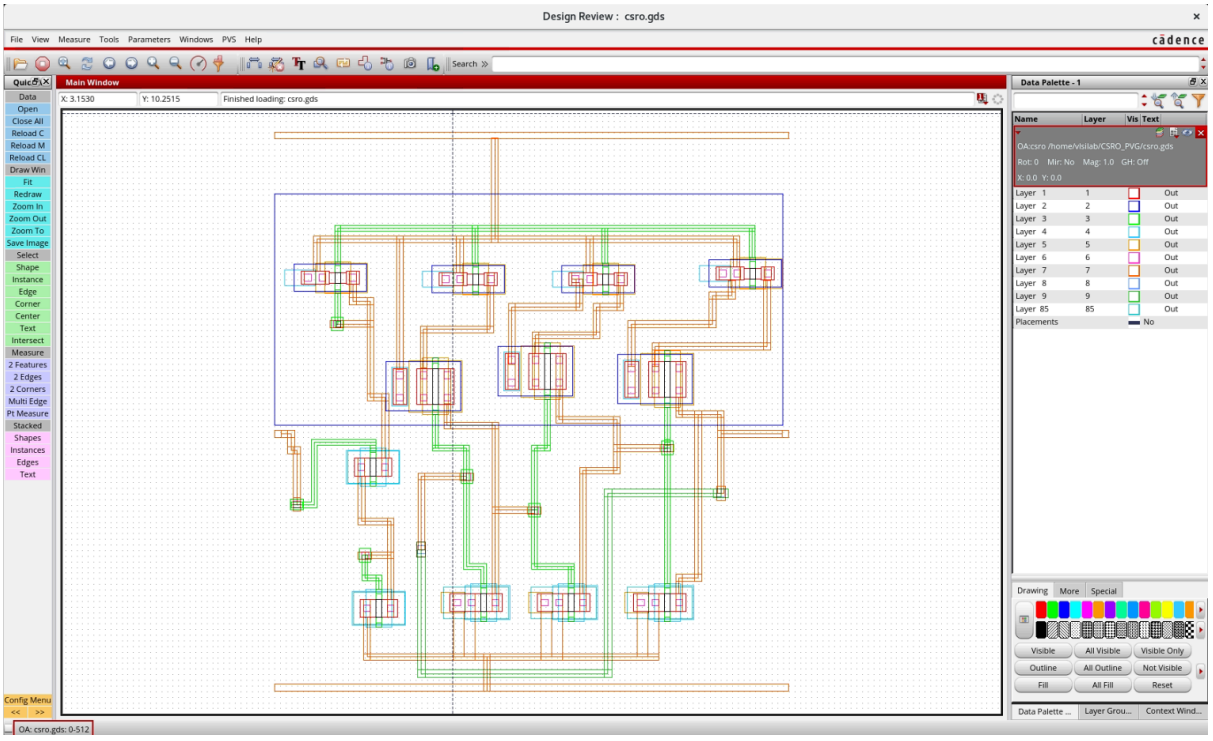

Run: "csro"
Run: "csro" from
/home/vlsilab/CSRO_PVG/
Schematic and Layout Match.
You currently have an open run (project).
Do you want to close current project and view the results of new run?
Summary of LVS Issues
Extraction Information:
0 cells have 0 mal-formed device problems
0 cells have 0 label short problems
0 cells have 0 label open problems
Comparison Information:
0 cells have 0 Net mismatches
0 cells have 0 Device mismatches
0 cells have 0 Pin mismatches
0 cells have 0 Parameter mismatches
ELW Information:
Total DRC violations: 0
Yes
No
Help


LVS Debug - csro
*** Schematic and Layout Match
Extract
Compare
Virtuoso® Layout Suite XL Editing: CSRO csro
Virtuoso® Schematic Editor L Reading: CSRO csro schematic Config: CSRO csro physCon

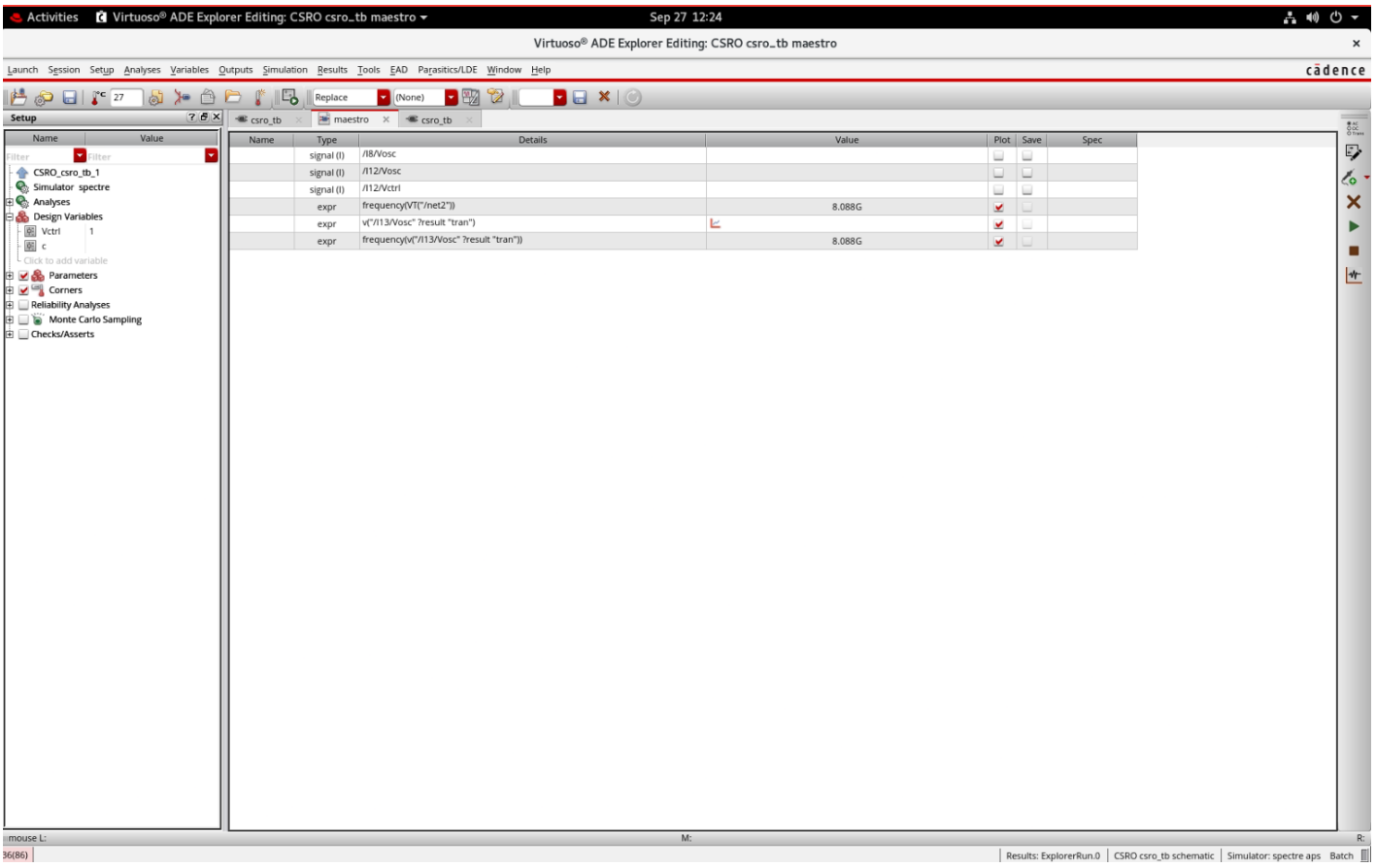
PEX



GDS FILE


Design Review : csro.gds
File View Measure Tools Parameters Windows PVS Help
cadence
Main Window
Finished loading: csro.gds
Data Palette - 1
Name Layer Vis Text
Drawing More Special
Visible All Visible Visible Only
Outline All Outline Not Visible
Fill All Fill Reset

Post-Layout Simulation Results



Post-layout simulation was performed using the extracted (av_extracted) view of the CSRO layout in Cadence Virtuoso. The extracted view includes the parasitic effects introduced by the physical layout and interconnects. At vdd =1.8 V and vctrl=1 V the oscillator exhibited sustained oscillation with a post-layout oscillation frequency of approximately 8.088 GHz. The pre-layout schematic simulation produced a frequency of approximately 8.076 GHz.

Thus, only a small variation in oscillation frequency was observed after layout extraction, indicating that the implemented layout preserves the intended oscillator operation.

10. Results and Analysis

The proposed **three-stage Current-Starved Ring Oscillator with PMOS current sources** was designed and simulated using **90 nm CMOS technology** in Cadence Virtuoso. The circuit was evaluated for its oscillation behavior, frequency tuning capability, power consumption, PVT variations, and statistical variations using Monte Carlo analysis.

The transient simulation confirms that the proposed ring oscillator successfully generates a continuous periodic output waveform. Since the oscillator consists of three inverter stages connected in a closed loop, the signal propagates through the stages and produces sustained oscillation at the output node.

At $\mathbf{V_{DD}}$ **= 1.8 V and** $\mathbf{V_{ctrl}}$ **= 1 V**, the nominal transient simulation produced an oscillation frequency of approximately **8.076 GHz**. The frequency was obtained in Cadence Virtuoso using the frequency measurement expression applied to the output waveform. This confirms the proper operation of the proposed three-stage oscillator.

The frequency-control analysis shows that the oscillation frequency increases with an increase in $\mathbf{V_{ctrl}}$. The control voltage regulates the current supplied to the inverter stages through the PMOS current-source arrangement. When Vctrl is increased, the available current increases, reducing the propagation delay of the stages and consequently increasing the oscillation frequency. Therefore, Vctrl can be used to tune the output frequency of the oscillator.

The simulation results also show a clear trade-off between **frequency and power consumption**. At higher control voltages, the oscillator operates at a higher frequency but consumes more power because of the increased current flow through the circuit. At lower control voltages, the available current is reduced, resulting in lower frequency and significantly reduced power consumption.

The PVT analysis was performed using the **FF, FS, NN, SF, and SS process corners** at temperatures of **-40°C, 27°C, and 125°C**. The results show that both frequency and power vary with process and temperature conditions. The simulated frequency ranges from approximately **1.238 GHz to 13.87 GHz**, while the power consumption ranges from approximately **19.24 µW to 469.9 µW** for the investigated PVT conditions.

Monte Carlo analysis was further performed using **200 samples** to evaluate the effect of random device variations. At $\mathbf{V_{ctrl}}$ **= 1 V**, the mean frequency obtained was approximately **7.96888 GHz**, with a standard deviation of **636.128 MHz**, while the mean power consumption was approximately **261.445 µW** with a standard deviation of **28.6619 µW**. At $\mathbf{V_{ctrl}}$ **= 0.1 V**, the mean frequency was approximately **420.938 MHz**, with a standard deviation of **132.900 MHz**, and the mean power consumption was approximately **3.49059 µW** with a standard deviation of **1.28092 µW**.

The Monte Carlo simulation achieved a **100% yield for the 200 simulated samples** for both control-voltage conditions. This indicates that the oscillator successfully produced valid results for all the simulated random variations under the selected simulation conditions.

11. Conclusion

The overall simulation results demonstrate that the proposed **90 nm PMOS-source current-starved ring oscillator** provides a useful combination of **high-frequency operation, frequency tunability, low-power operation, and robustness against process and temperature variations**. The circuit can therefore serve as a suitable oscillator architecture for applications where controllable frequency and power efficiency are important.